



Notice

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Physical Property Table
Data shown are typical values obtained by proper testing methods and should not be used for specification purpose.
Please use these data for selecting the most appropriate grade suitable for specific usage. These data may be changed because of the improvement in properties.

Important Precautions and Notes on Handling and Use
Always observe the following general precautions in the handling and use of XYRON™, and consult the material safety data sheet, which will be made available upon request, before handling and use. Prior to mixing or using any additive or other substance with XYRON™, investigate and determine its safety and suitability for that purpose.

Health and Safety
Avoid exposure of skin, eyes, and respiratory tract to gas generated during melting and drying. Avoid contact exposure to hot or molten resin. Employ local ventilation and protective gear, including chemical goggles or safety glasses, and protective gloves, throughout drying and melting operations. In case of emergency, call one of the following:US: +1-800-424-9300 (Chemtrec) ,EU: +32-1-458-4545 (BIG v.z.w.) ,Japan: +81-44-271-2561 (Asahi Kasei Corporation)

Combustibility
XYRON™ is combustible and must be kept strictly away from heat, sparks, and flame during handling and storage. XYRON™ may emit toxic gases during combustion. Extinguish with water, or with foam or powder chemical extinguisher.

Disposal
Dispose of XYRON™ by landfill burial or by incineration in a properly equipped facility, in accordance with relevant laws and regulations. XYRON™ may emit carbon monoxide and other toxic gasses during incineration.

Storing
Keep XYRON™ strictly away from direct sunlight, water, high humidity, and moisture.

Proscribed Applications
Do not use XYRON™ for food packaging applications.
Do not use XYRON™ for medical devices designed or intended for intracorporeal implantation or contact, whether direct or indirect, with bodily fluids.
For specifics, refer to our division. Asahi Kasei Corporation encourages its customers to review their applications of its products from the standpoint of human health and environmental quality. To help ensure that Asahi Kasei Corporation products are not used in ways for which they are not intended or tested, Asahi Kasei Corporation representatives are willing to assist in dealing with ecological and product safety considerations.

Compliance with Relevant Standards
XYRON™ grades are available which comply with Japan's "voluntary industrial guidelines on polyolefin and other synthetic resins for use as food container or food wrapping materials," with Notification No.370 of the Ministry of Health and Welfare of Japan, and with UL and CSA standards. In all cases, it is nevertheless recommended that the safety of XYRON™ for a specific application be confirmed by the user through appropriate testing. For further information, please contact Asahi Kasei Corporation.

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Modified Polyphenylene Ether Resin

XYRON™

Physical Properties

Material that expands possibilities

Asahi**KASEI**

XYRON™ (modified polyphenylene ether or mPPE) engineering resins are composed of PPE together with polystyrene, polyamide, polypropylene, polyphenylene sulfide or other polymeric materials.

Major Advantages

1

Heat resistance (90-220 degrees C)

2

Halogen-free flame retardancy (UL 94 V-0 through HB)

3

Excellent electrical properties

4

-Low mold shrinkage
-Excellent dimensional accuracy and stability

5

-Low water absorption
-Hydrolytic stability
-Hot water resistance

6

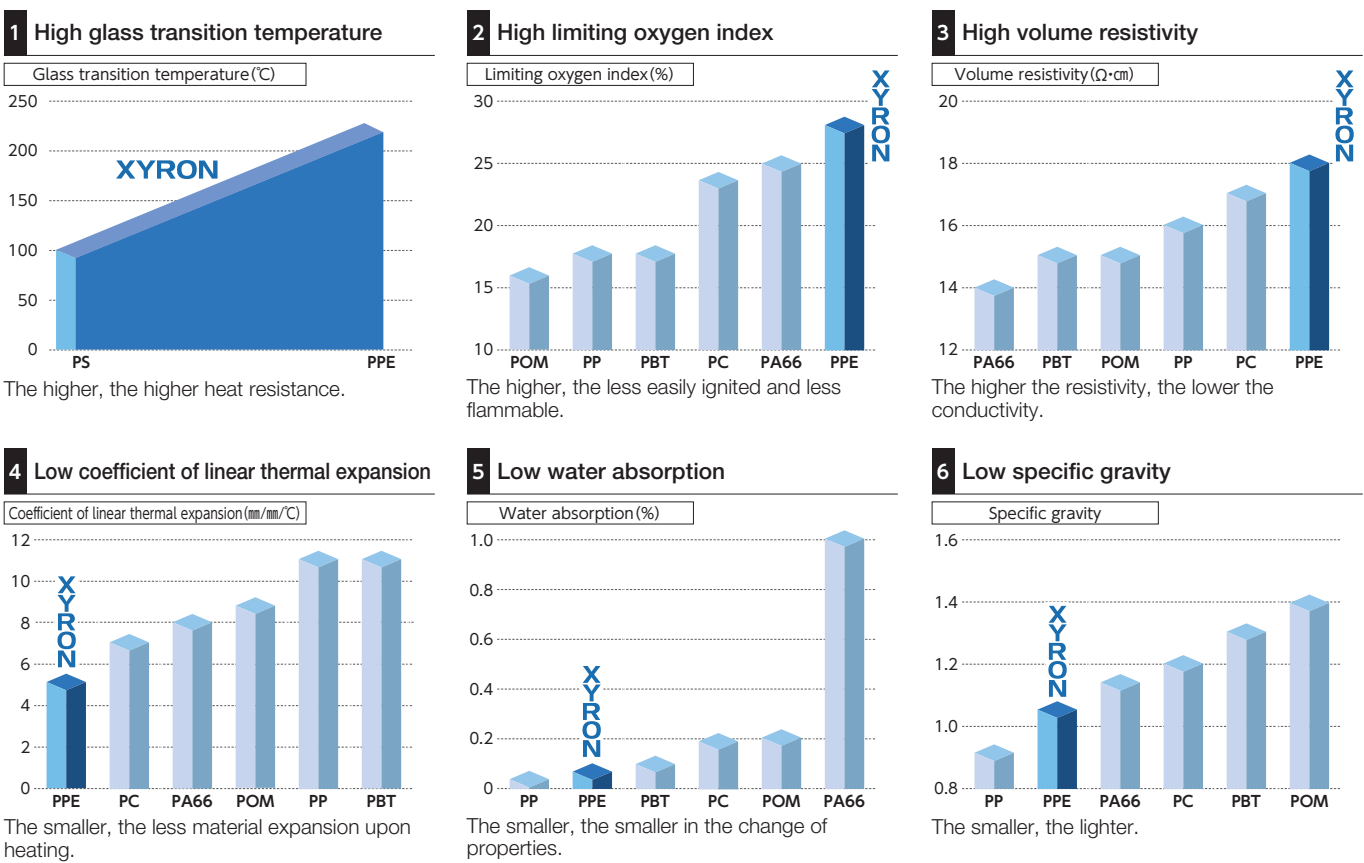
Low specific gravity

7

Resistance to acids and alkalis

8

-Processability
-Reworkability



XYRON™ Grades (PPE/PS)

	ASTM Method	ISO Method	Flammability (UL 94)			
	HDT(°C) 1.82MPa	DTUL(°C) 1.8MPa	HB	V-2	V-1	V-0
Non-Reinforced	76	70~77	-	-	1951J	-
	85	78	-	-	100V	100Z
	90	86~93	200H	240W	240V	240Z
	100	95~98	300H	340W	340V	340Z
	110	104~109	400H	-	-	440Z
	120	112~114	500H	-	540V·CN51V	540Z
	130	126	600H	-	640V	644Z
	140	135~137	-	-	740V	744Z
	150	146	-	-	-	SZ800
	160	-	-	-	-	-
Reinforced	170	158	1000H(Eq. to HB)	-	-	-

I XYRON™ Grade List

					H B				Flame Retardant														
Property		Units	Test method	Test Condition	200H	300H	500H	1000H	1951J	240V	240Z	340V	340Z	440Z	443Z	540V	540Z	644Z	740V	744Z	SZ800	CN51V	
Physical	Specific Gravity		JIS K 7112	23℃	1.05	1.05	1.06	1.07	1.11	1.09	1.10	1.09	1.10	1.10	1.10	1.09	1.10	1.10	1.09	1.10	1.10	1.09	
	Moisture Absorption	%	ISO 62	Immersion for 24 hours	0.06	0.06	0.06	0.06	0.10	0.10	0.10	0.10	0.10	0.10	-	0.10	0.10	0.10	0.10	-	0.08	-	
Thermal	Flammability		UL 94	1.6mm	HB	HB	HB	Eq. to HB	V-1*/2.0mm	V-1	V-0*	V-1	V-0*	V-0*	V-0*	V-1	V-0*	V-0*	V-1	V-0	V-0	V-1	
	Temperature of Deflection Under Load(DTUL)	℃	ASTM D 648	1.82MPa	90	100	120	170	76	90	90	100	100	110	-	120	120	130	140	-	150	-	
			ISO 75-1	1.8MPa	93	98	114	158	70	88	86	95	96	104	104	114	112	126	137	135	146	111	
	Coefficient of Linear Thermal Expansion	X10 ⁻⁵ mm/mm/℃	ISO 11359	-30℃ 65℃	MD	7.4	6.7	6.8	6.0	7.4	6.8	6.8	6.7	6.7	6.5	-	5.8	6.0	6.1	6.2	6.5	6.0	7.3
				TD	7.5	7.9	7.2	6.2	8.1	7.6	7.3	7.2	7.1	7.2	-	6.4	6.9	7.2	6.4	6.6	6.2	8.1	
	Moulding Shrinkage	%	ASAHIKASEI	150×150×2mm	0.5~0.7	0.5~0.7	0.6~0.8	0.7~0.8	0.6~0.7	0.5~0.7	0.5~0.7	0.6~0.7	0.6~0.8	0.6~0.8	0.6~0.8	0.6~0.8	0.6~0.8	0.7~0.9	0.6~0.7	0.8~1.0	0.8~1.0	0.5~0.7	
Electrical	Dielectric Constant		IEC 60250	100Hz	2.8	2.8	2.8	2.8	2.9	2.9	2.9	2.9	2.9	2.9	-	2.9	2.9	2.9	2.9	-	2.8	-	
			SPDR	5GHz	2.6	2.6	2.6	2.6	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.6	2.7	2.7	2.6	
	Dissipation Factor		IEC 60250	100Hz	0.0005	0.0005	0.0005	0.0005	0.0030	0.0030	0.0030	0.0030	0.0030	0.0030	-	0.0030	0.0030	0.0040	0.0030	-	0.0027	-	
			SPDR	5GHz	0.002	0.002	0.001	0.002	0.004	0.004	0.005	0.005	0.004	0.004	0.004	0.005	0.004	0.005	0.003	0.004	0.005	0.004	
	Volume Resistivity	Ω·cm	IEC 60093	23℃ 50% RH	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	-	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	
	Surface Resistivity	Ω			10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	-	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶		
Mechanical	Tensile Strength	MPa	ISO 527	23℃ 50% RH	41	45	53	70	43	52	60	52	59	67	62	65	69	74	68	75	72	66	
	(Nominal)Tensile Strain	%			35	31	31	24	24	17	9	14	13	8	15	13	12	9	11	9	19	15	
	Flexural Strength	MPa	ISO 178		71	79	92	112	68	85	95	93	102	106	96	102	107	112	101	111	110	100	
	Flexural Modulus				2400	2400	2300	2400	2300	2500	2700	2500	2500	2700	2270	2500	2500	2600	2500	2400	2400	2400	
	Charpy Impact Strength	kJ/m ²	ISO 179		4mm 23℃(Notched)	13	18	19	8	13	17	16	18	16	17	40	18	19	20	15	16	13	28
Molding Conditions	Resin Temperature	℃	-	-	220~270	240~280	240~290	280~320	220~280	220~270	220~270	240~280	240~280	240~290	290~320	240~300	240~300	260~320	250~320	250~320	250~320	240~290	
	Mold Temperature				40~70	50~80	50~80	80~120	40~70	40~70	40~70	50~80	50~80	50~80	60~100	50~80	50~80	60~100	60~100	60~100	80~120	50~80	
	Pre-Drying Temperature				80~90	90~100	90~100	100~120	70~80	80~90	80~90	90~100	90~100	90~100	90~100	90~100	90~100	90~100	90~100	90~100	100~120	90~100	
	Pre-Drying Time	Hr	-	-	2~4	2~4	2~4	2~4	2~4	2~4	2~4	2~4	2~4	2~4	2~4	2~4	2~4	2~4	2~4	2~4	2~4		
Remark					-	-	-	-	*5VB/2.0mm	-	*5VA/2.5mm	-	*5VA/2.5mm	*5VA/2.5mm	*5VA/2.5mm	-	*5VA/2.5mm *(f1)BK	*5VA/2.0mm *(f1)BK RTI:125℃	-	-	-	Light transmission grade: Wavelength 550 nm: 51% transmission(2 mm t)	

					PA / PPE						PP / PPE						PPS / PPE				PPA / PPE
Property		Units	Test method	Test Condition	A0210	A0230	AT600	AG511	AG213	AF700	TZ100	TF701	TA720	T0703	TG102	TT521	DG235	DG040	DG141	DV166	XP640
Physical	Specific Gravity		JIS K 7112	23℃	1.11	1.10	1.09	1.17	1.33	1.12	0.93	1.07	1.21	1.01	1.18	1.13	1.44	1.52	1.55	1.82	1.45
	Moisture Absorption	%	ISO 62	Immersion for 24 hours	—	—	—	—	—	—	—	—	—	—	—	0.04	—	—	—	—	0.13
Thermal	Flammability		UL 94	1.6mm	—	—	HB	—	—	V-0	—	V-0	Eq. to V-0	HB	Eq. to HB	HB	V-1	V-1	V-0*/2.0mm	V-1/2.5mm	HB
	Temperature of Deflection Under Load(DTUL)	℃	ASTM D 648	1.82MPa	—	—	—	—	—	—	97*	—	—	—	—	120*	—	—	—	—	—
			ISO 75-1	1.8MPa	176*	179*	190*	198*	252*	193*	93*	107*	106*	118*	138*	109*	217	183	214	256	274
	Coefficient of Linear Thermal Expansion	X10 ⁻⁵ mm/mm/℃	ISO 11359	-30℃ ~ 65℃	MD	8.5	—	9.0	4.0	2.3	5.4	—	—	—	1.8	5.1	2.1	2.2	1.9	1.4	1.4
				TD	9.3	—	9.0	9.9	8.9	8.8	9.7	—	—	—	—	14	7.6	5.3	3.3	4.1	3.1
	Moulding Shrinkage	%	ASAHIKASEI	150×150×2mm	1.3~1.3	—	1.4~1.4	0.6~0.7	0.3~1.0	1.5~1.6	1.8~2.0	0.8~1.2	0.7~0.9	1.0~1.2	0.28~0.67	0.7~0.9	0.2~0.7	0.2~0.3	0.2~0.5	0.1~0.2	0.3~0.8
Electrical	Dielectric Constant		IEC 60250	100Hz	3.2	—	3.1	3.3	3.5	—	2.3	—	—	—	—	2.5	4.0	—	4.1	4.8	—
			SPDR	5GHz	2.9	—	2.8	3.0	—	2.9	2.3	2.7	—	2.5	2.9	—	3.5	—	3.8	—	3.7
	Dissipation Factor		IEC 60250	100Hz	0.007	—	0.005	0.006	0.005	—	0.0002	—	—	—	—	0.0002	0.0010	—	0.0010	0.0210	—
			SPDR	5GHz	0.011	—	0.008	0.010	—	0.010	0.001	0.006	—	0.001	0.003	—	0.005	—	0.005	—	0.009
	Volume Resistivity	Ω•cm	IEC 60093	23℃ 50% RH	2.1×10 ¹⁵	—	4.1×10 ¹⁵	1.3×10 ¹⁶	4.7×10 ¹⁵	—	8.4×10 ¹⁶	—	—	—	—	5.5×10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	—
	Surface Resistivity	Ω			—	—	—	—	—	—	—	—	—	—	—	—	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	—
Mechanical	Tensile Strength	MPa	ISO 527	23℃ 50% RH	65	65	61	89	152	54	36	53	43	43	78	41	122	66	88	87	173
	(Nominal)Tensile Strain	%			42	33	54	3	3	44	60	12	4	16	4	20	2	1	1	1	2
	Flexural Strength	MPa	ISO 178		97	95	92	139	238	92	55	77	78	60	130	62	187	106	144	143	257
	Flexural Modulus				2500	2300	2400	4200	8700	2700	2200	2270	3700	1900	7100	2900	10200	9500	12800	18100	12500
	Charpy Impact Strength	kJ/m ²	ISO 179		4mm 23℃(Notched)	21	24	30	10	11	6	26	3	3	15	14	6	5	2	3	6
Molding Conditions	Resin Temperature	℃	—	—	280~300	280~300	280~300	260~300	280~300	270~290	230~260	230~270	250~290	250~290	250~290	250~280	300~330	300~330	300~330	300~330	320~340
	Mold Temperature				60~120	60~120	60~120	60~120	60~120	60~120	50~70	50~80	50~90	50~90	50~90	50~90	120~150	120~150	120~150	120~150	130~150
	Pre-Drying Temperature				110~130	110~130	110~130	100~130	110~130	110~130	80~90	90~100	90~100	90~100	90~110	90~110	120	120	120	120	110~120
	Pre-Drying Time	Hr	—	—	2~3	2~3	2~4	3~4	2~4	2~4	1~2	2~4	2~4	2~4	2~4	3~4	3~4	3~4	3~4	3~4	3~4
Remark					*Low load (0.45MPa)	*Low load (0.45MPa)	*Low load (0.45MPa)	GF 10% , Good appearance *Low load (0.45MPa)	GF 30% *Low load (0.45MPa)	*Low load (0.45MPa)	*Annealing	*Annealing	Filler 20% *Annealing	*Annealing	GF 30% *Annealing	Filler 20% *Annealing	GF 30%	Filler 40%	Filler 40% *5VB/2.0mm *V-1/0.75mm *(f1)BK	Filler 60%	GF 40%

I XYRON™ Grade List

					Glass Fiber				Flame Retardant, Glass Fiber				High-Stiffness•Low-Warp								
Property		Units	Test method	Test Condition	G701H	G402H	G702H	G703H	G601Z	G701V	G702V	G793Z	X332V	X533H	X423V	X444V	X7744	G544V	M544V	X523V	
Physical	Specific Gravity		JIS K 7112	23℃	1.12	1.19	1.21	1.28	1.17	1.15	1.23	1.35	1.23	1.30	1.32	1.39	1.39	1.40	1.38	1.34	
	Moisture Absorption	%	ISO 62	Immersion for 24 hours	0.06	—	0.06	0.06	—	0.06	0.06	—	0.06	—	—	—	—	—	—	—	
Thermal	Flammability		UL 94	1.6mm	HB	HB	HB	HB	V-0	V-1*	V-1	V-0	V-1	HB	V-1	V-1	V-1	V-1	Eq. to V-1	V-1	
	Temperature of Deflection Under Load(DTUL)	℃	ASTM D 648	1.82MPa	130	—	140	140	—	130	140	—	100	—	—	—	—	—	—	—	
			ISO 75-1	1.8MPa	124	113	136	135	118	130	136	148	104	121	106	116	116	113	119	118	
	Coefficient of Linear Thermal Expansion	X10 ⁻⁵ mm/mm/℃	ISO 11359	-30℃	MD	4.1	3.2	2.7	2.2	3.8	3.7	2.5	2.1	3.2	2.9	1.9	2.7	2.0	2.1	2.2	3.1
				65℃	TD	7.3	7.5	6.9	6.9	7.3	6.5	6.5	6.3	6.3	5.9	6.0	5.6	5.4	5.7	6.1	6.2
	Moulding Shrinkage	%	ASAHIKASEI	150×150×2mm	0.50~0.55	0.23~0.50	0.26~0.54	0.20~0.50	0.48~0.49	0.50~0.50	0.27~0.56	0.19~0.69	0.27~0.42	0.19~0.40	0.19~0.33	0.13~0.24	0.13~0.41	0.14~0.38	0.13~0.35	0.19~0.38	
Electrical	Dielectric Constant		IEC 60250	100Hz	2.9	—	3.1	3.3	—	3.0	3.2	—	3.2	—	—	—	—	—	—	3.3	
			SPDR	5GHz	2.7	2.9	3.0	—	—	2.8	3.0	3.3	3.0	3.2	—	3.4	3.4	3.4	—	—	
	Dissipation Factor		IEC 60250	100Hz	0.0004	—	0.0006	0.0009	—	0.0020	0.0030	—	0.0040	—	—	—	—	—	—	0.0060	
			SPDR	5GHz	0.002	0.003	0.003	—	—	0.004	0.004	0.008	0.005	0.003	—	0.005	0.006	0.006	—	—	
	Volume Resistivity	Ω•cm	IEC 60093	23℃ 50% RH	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	—	10 ¹⁶
	Surface Resistivity	Ω			10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	—	10 ¹⁶	
Mechanical	Tensile Strength	MPa	ISO 527	23℃ 50% RH	74	95	105	125	84	88	113	127	72	79	84	88	98	104	111	60	
	(Nominal)Tensile Strain	%			3	2	2	2	4	3	2	2	2	2	2	2	1	2	2	3	
	Flexural Strength	MPa	ISO 178		130	158	164	198	136	138	162	189	120	140	132	140	160	175	188	106	
	Flexural Modulus				4300	6900	6500	9300	3700	4600	6700	9100	6600	8600	7300	9900	10900	10000	10800	6200	
	Charpy Impact Strength	kJ/m ²	ISO 179		4mm 23℃(Notched)	10	11	10	10	9	7	8	11	5	6	5	6	6	6	6	3
Molding Conditions	Resin Temperature	℃	—	—	260~300	240~300	260~300	260~300	260~300	260~300	260~300	260~300	240~300	250~300	240~300	250~300	250~300	250~300	250~300	250~300	
	Mold Temperature				60~100	50~80	60~100	60~100	60~100	60~100	60~100	60~100	50~80	70~90	50~80	70~90	70~90	70~90	70~90	70~90	
	Pre-Drying Temperature				90~100	90~100	90~100	90~100	90~100	90~100	90~100	90~100	90~100	90~100	90~100	90~100	90~100	90~100	90~100	90~100	
	Pre-Drying Time	Hr	—	—	2~4	2~4	2~4	2~4	2~4	2~4	2~4	2~4	2~4	2~4	2~4	2~4	2~4	2~4	2~4	2~4	
Remark					GF 10%	GF 20%	GF 20%	GF 30%	GF 10%	GF 10% *5VA/2.4mm	GF 20%	GF 30%	Filler 20%	Filler 30%	Filler 30%	Filler 35%	Filler 35%	Filler 35%	Filler 35%	Filler 30%	

					High-Stiffness•Ultra Low-Warp			Good Appearance, Low-Warp			Electroconductive•High-Stiffness		Acoustic Damping					Lubricated	Drinking Water Standard Compliance		
Property		Units	Test method	Test Condition	X364V	X1712	M665V	X552H	X352V	X552V	X8610	C443V	VN30V	VT31V	VTD1V	VM34V	VT52Z	FG70H	W5002	WG722	WG732
Physical	Specific Gravity		JIS K 7112	23℃	1.39	1.45	1.56	1.20	1.23	1.23	1.16	1.33	1.08	1.15	1.13	1.38	1.25	1.18	1.06	1.20	1.28
	Moisture Absorption	%	ISO 62	Immersion for 24 hours	–	0.06	–	0.06	0.06	0.06	0.06	0.06	0.06	0.06	–	–	0.06	–	0.06	0.06	0.06
Thermal	Flammability		UL 94	1.6mm	V-1	V-0	Eq. to V-1	HB	V-1	V-1*	V-1	V-1	V-1	V-1	Eq. to V-1	V-1	V-0	Eq. to HB	Eq. to HB	Eq. to HB	Eq. to HB
	Temperature of Deflection Under Load(DTUL)	℃	ASTM D 648	1.82MPa	–	110	–	120	100	120	–	–	100	100	–	–	–	–	–	–	–
			ISO 75-1	1.8MPa	107	109	129	115	100	117	123	110	96	95	98	105	113	141	114	144	144
	Coefficient of Linear Thermal Expansion	X10 ⁻⁵ mm/mm/℃	ISO 11359	-30℃ MD	2.8	2.6	1.9	3.9	3.8	3.7	1.7	1.4	6.5	5.1	6.2	2.1	3.5	–	7.0	2.5	2.3
				65℃ TD	4.2	4.4	3.4	5.6	5.3	5.2	7.3	5.3	7.4	7.0	8.7	5.6	5.7	–	8.1	6.7	6.4
	Moulding Shrinkage	%	ASAHIKASEI	150×150×2mm	0.13~0.17	0.14~0.22	0.13~0.18	0.36~0.47	0.30~0.40	0.33~0.44	0.19~0.45	0.10~0.40	0.50~0.60	0.40~0.50	0.43~0.58	0.18~0.32	0.37~0.44	0.3~0.6	0.60~0.80	0.26~0.54	0.20~0.50
Electrical	Dielectric Constant		IEC 60250	100Hz	–	3.4	–	3.2	3.2	3.2	–	–	–	–	–	–	–	–	–	–	–
			SPDR	5GHz	3.3	3.6	–	2.9	2.9	2.9	–	–	2.6	2.8	2.7	–	2.9	–	2.6	2.9	3.1
	Dissipation Factor		IEC 60250	100Hz	–	0.0060	–	0.0030	0.0030	0.0030	–	–	–	–	–	–	–	–	–	–	–
			SPDR	5GHz	0.004	0.009	–	0.003	0.004	0.005	–	–	0.003	0.004	0.004	–	0.005	–	0.002	0.003	0.003
	Volume Resistivity	Ω•cm	IEC 60093	23℃ 50% RH	10 ¹⁶	10 ¹⁶	–	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ²⁻³	10 ²⁻³	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	–	10 ¹⁶	10 ¹⁶	10 ¹⁶
	Surface Resistivity	Ω			10 ¹⁶	10 ¹⁶	–	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ²⁻³	10 ²⁻³	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	10 ¹⁶	–	10 ¹⁶	10 ¹⁶	10 ¹⁶
Mechanical	Tensile Strength	MPa	ISO 527	23℃ 50% RH	59	68	60	55	53	62	90	73	45	44	46	63	54	83	63	121	140
	(Nominal)Tensile Strain	%			3	1	1	6	8	5	1	1	24	43	21	3	13	2	7	2	2
	Flexural Strength	MPa	ISO 178		101	118	102	109	102	117	130	117	74	75	74	103	100	120	100	187	216
	Flexural Modulus				9200	9200	13600	5300	5000	5300	8500	12000	1900	2400	2600	5600	4100	5700	2600	6900	8900
	Charpy Impact Strength	kJ/m ²	ISO 179		4mm 23℃(Notched)	2	4	2	4	6	4	7	4	4	9	20	7	5	9	20	12
Molding Conditions	Resin Temperature	℃	–	–	260~310	250~300	270~310	250~300	240~300	250~300	250~300	250~300	250~290	250~290	250~290	250~290	260~300	240~300	240~290	260~300	260~300
	Mold Temperature				60~100	60~90	60~120	70~90	50~80	70~90	60~90	60~90	50~80	50~80	50~80	50~80	60~90	60~100	50~80	60~100	60~100
	Pre-Drying Temperature				90~100	90~100	90~110	100~110	100~110	100~110	90~100	90~100	90~100	100~110	100~110	100~110	100~110	90~100	90~100	90~100	90~100
	Pre-Drying Time	Hr	–	–	2~4	2~4	2~4	3~4	3~4	3~4	2~4	2~4	2~4	3~4	3~4	3~4	3~4	2~4	2~4	2~4	2~4
Remark					Filler 35%	Filler 40%	Filler 50%	Filler 20%	Filler 20%	Filler 20% *5VA/2.5mm	Filler 10%	Filler 30%	Loss factor 6.6	Filler 10% Loss factor 9.5	Filler 10% Loss factor 6.3	Filler 35% Loss factor 1.3	Filler 20% Loss factor 6.7	–	NSF.KTW, W270.ACS, WRAS	Filler 20% NSF.KTW, W270.ACS, WRAS	Filler 30% NSF.KTW, W270.ACS, WRAS